

# Unit 3: Practical Skills in Chemistry I

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## IAS compulsory unit

## Externally assessed

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### Unit description

#### Introduction

This unit consists of a written practical examination, covering the skills and techniques developed during practical work in Units 1 and 2.

Although the unit content contains eight core practical activities, the examination will not be limited to recall of these core practicals but may include questions where students are expected to apply their knowledge to new practical situations.

Students should, therefore, develop their practical skills by completing a range of different practicals that require a variety of different techniques.

As students carry out practical activities, they should be encouraged to write laboratory reports using appropriate scientific, technical and mathematical language, conventions and symbols.

#### Development of practical skills, knowledge and understanding

Students are expected to develop experimental skills and a knowledge and understanding of the necessary techniques by carrying out a range of practicals while they study Units 1 and 2.

This unit will assess students' knowledge and understanding of the practical procedures and techniques they develop.

To prepare for assessment of this unit, centres should give students opportunities to carry out practical activities, collect and analyse data, and draw conclusions. Students should – at the least – carry out the eight core practicals in class. By completing these practicals students will be able to:

- follow and interpret experimental instructions, covering the full range of laboratory exercises set throughout the course, with minimal help from the teacher
  - always work with interest and enthusiasm in the laboratory, completing most laboratory exercises in the time allocated
  - manipulate apparatus, use chemicals, carry out all common laboratory procedures and use data logging (where appropriate) with the highest level of skill that may be reasonably expected at this level
  - work sensibly and safely in the laboratory, paying due regard to health and safety requirements without the need for reminders from the teacher
  - gain accurate and consistent results in quantitative exercises, make most of the expected observations in qualitative exercises and obtain products in preparations of high yield and purity.
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## Assessment information

- First assessment: June 2019.
  - The assessment is 1 hour and 20 minutes.
  - The assessment is out of 50 marks.
  - Students must answer all questions.
  - This paper may include short-open, open-response and calculation questions.
  - This paper will include a minimum of 6 marks that target mathematics at Level 2 or above.
  - Students will be expected to apply their knowledge and understanding of practical skills to familiar and unfamiliar situations.
  - Calculators may be used in the examination (see *Appendix 8: Use of calculators*).
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## **The question paper will assess the following abilities.**

### **Independent thinking in a practical context**

- Solve problems set in a practical context.
- Apply scientific knowledge to practical contexts.

### **Use and application of scientific methods and practices**

- Identify and state how to control variables to improve experimental validity.
- Present data in appropriate ways.
- Evaluate results and draw conclusions.
- Appreciate measurement uncertainties and errors.
- Comment on the method for an experiment.

### **Numeracy and the application of mathematical concepts in a practical context**

- Plot and interpret graphs.
- Process and analyse data using appropriate mathematical skills.
- Use appropriate numbers of significant figures based on the experimental data.
- Consider the accuracy and precision of data.

### **Use of apparatus and equipment**

- Recognise a range of laboratory apparatus and select appropriate apparatus for a particular scenario.
- Understand how to use a range of apparatus and techniques appropriate to the knowledge and understanding included in this specification.
- Consider the range and resolution of apparatus.
- Identify health and safety issues and discuss how these may be dealt with.

## **Questions may require students to:**

- recall and/or interpret observations relating to tests for ions and gases in Units 1 and 2
  - recall and/or interpret observations relating to tests for organic functional groups in Units 1 and 2
  - manipulate data and comment on experimental methods and techniques for a range of experiments involving measurements in Units 1 and 2, including molar mass calculations, titrations, thermochemical investigations and simple kinetics experiments
  - comment on experimental methods and techniques in the preparation of inorganic or organic compounds in Units 1 and 2.
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# Unit 4: Rates, Equilibria and Further Organic Chemistry

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## IA2 compulsory unit

## Externally assessed

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### Unit description

#### Introduction

In this unit, students make a quantitative study of chemical kinetics and extend their study of organic reaction mechanisms.

The topics of entropy and equilibria show how chemists are able to predict quantitatively the direction and extent of chemical change.

The unit tests the equilibrium law by showing the degree to which it can accurately predict changes during acid-base reactions, notably the changes to pH during titrations.

The organic chemistry in this unit covers carbonyl compounds, and carboxylic acids and their derivatives.

Students are required to apply their knowledge gained in Units 1 and 2, to all aspects of this unit. This includes nomenclature, ideas of isomerism, bond polarity and bond enthalpy, reagents and reaction conditions, reaction types and mechanisms. Students are also expected to use formulae and balanced equations and calculate chemical quantities.

#### Chemistry in action

This unit shows how the principles of kinetics and thermodynamics can help to identify optimal conditions for the manufacture of chemicals.

The study of entropy links thermodynamics and equilibrium, and shows how chemists approach fundamental questions about the stability of chemicals and the direction of chemical change.

The historical development of theories explaining acids and bases shows how scientific ideas change as a result of new evidence and fresh thinking.

The study of buffer solutions shows the importance of equilibrium systems in living cells, in medicines, in foods and in the natural environment.

The two broad areas of application of chemistry are synthesis and analysis. In this unit, synthesis is illustrated by reactions of carbonyl compounds (notably with cyanide ions) and the production of esters for use as solvents, flavourings and perfumes. The main analytical technique featured is nuclear magnetic resonance (NMR), including coverage of magnetic resonance imaging.

## Practical skills

Through practical work, students will learn about the methods used to measure reaction rates. They will collect data, analyse it and interpret the results. They then see how knowledge of rate equations and other evidence can enable chemists to propose models to describe the mechanisms of reactions.

Simple practical work can be used to investigate equilibrium systems.

Students can develop their skill at volumetric analysis, with a number of titration activities as part of their exploration of acid-base equilibria.

Although the organic section of the unit contains no core practical activities, students would be expected to encounter simple test-tube reactions for organic functional groups – such as the use of Benedict's or Tollens' reagents. There are also opportunities to undertake synthetic reactions, for example to make and purify an ester.

## Mathematical skills

There are opportunities for the development of mathematical skills in this unit. This includes plotting and justifying the shapes of rate-concentration and concentration-time graphs, calculating the half-life of a reaction, calculating the activation energy from a suitable graph and rearranging the Arrhenius equation. Also, calculating entropy changes, constructing Born-Haber cycles and calculating missing values, constructing expressions for  $K_c$  and  $K_p$  and calculating values with relevant units, estimating the change in value of an equilibrium constant when a variable changes, using logarithms and exponentials for converting from concentration to pH of a buffer solution, plotting and interpreting titration curves and representing chiral molecules with appropriate diagrams. There is also an opportunity for calculating  $R_f$  values and interpreting infrared spectra and using the  $(n + 1)$  rule for proton NMR. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

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## Assessment information

- First assessment: January 2020.
  - The assessment is 1 hour and 45 minutes.
  - The assessment is out of 90 marks.
  - Students must answer all questions.
  - This paper has three sections:
    - Section A: multiple choice questions
    - Section B: mixture of short-open, open-response, calculations and extended-writing questions
    - Section C: data or calculation question.
  - This paper will contain questions that require information from the Data Booklet (see *Appendix 9*).
  - This paper will include a minimum of 22 marks that target mathematics at Level 2 or above.
  - Students will be expected to apply their knowledge and understanding of experimental methods in familiar and unfamiliar contexts.
  - This paper may contain some synoptic questions which require knowledge and understanding from Units 1 and 2.
  - Calculators may be used in the examination (see *Appendix 8: Use of calculators*).
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